

The Markov Group Conjecture for Triangular Countable-State Processes

Candidate partial result for arXiv:2004.05995

11 August 2026

Status. Candidate partial result, likely valid. This note proves the triangular, or one-way acyclic, subcase and does not resolve the full Markov group conjecture.

1 Source conjecture

Glueck restates the Markov group conjecture as Conjecture 1.1 of [1], PDF page 1:

Let $T = (T_t)_{t \in [0, \infty)}$ be a Markovian C_0 -semigroup on ℓ^1 and assume that $T_1 : \ell^1 \rightarrow \ell^1$ is bijective (i.e., T extends to a C_0 -group). Then T has bounded generator.

Here Markovian means that every T_t is positive and norm-preserving on the positive cone. Figure 1 is a crop of the source page.

1. INTRODUCTION

The Markov group conjecture and its decoupled version. In 1967, the following problem was posed and partially analysed by Kendall [11] and Speakman [19].

Conjecture 1.1 (Markov group conjecture). *Let $T = (T_t)_{t \in [0, \infty)}$ be a Markovian C_0 -semigroup on ℓ^1 and assume that $T_1 : \ell^1 \rightarrow \ell^1$ is bijective (i.e., T extends to a C_0 -group). Then T has bounded generator.*

Here, *Markovian* (or *Markov*) means that, for each $t \geq 0$, the operator T_t is positive (in the sense that $T_t x \geq 0$ for all $x \geq 0$) and norm-preserving on the positive cone.

Figure 1: Conjecture 1.1 in [1, p. 1].

The source proves the conjecture for direct sums of finite-dimensional processes, with a dimension-free bound, but explicitly leaves open the passage from those decoupled systems to a general process.

2 Triangular Markov semigroups

Let $(e_j)_{j \in \mathbb{N}}$ be the canonical basis of $\ell^1(\mathbb{N})$. A Markov operator T has a nonnegative column-stochastic matrix

$$p_{ij} = \langle T e_j, e_i^* \rangle, \quad \sum_i p_{ij} = 1.$$

We call a Markov semigroup (T_t) *triangular* if, for every $t \geq 0$,

$$p_{ij}(t) = 0 \quad \text{whenever } i < j. \quad (1)$$

Thus mass can move from state j only to states $i \geq j$. A simultaneous permutation of the states merely replaces the displayed order by another fixed enumeration.

This class includes pure-birth and more general one-way processes with jumps of arbitrary length. It need not decompose into finite invariant blocks.

3 Idea of the proof

For a triangular process, a path that starts at state j and is at j at a later time could not have visited a larger state and returned. Algebraically, the semigroup product has only one contribution to its (j, j) entry. Hence the diagonal probabilities are exponentials $e^{-q_j t}$.

The inverse bound at time one controls these exponentials. Indeed, the first j coordinate functionals form a finite-dimensional invariant subspace of T_1^* , and its triangular matrix has $p_{jj}(1)$ as an eigenvalue. Thus $p_{jj}(1) \geq \|T_1^{-1}\|^{-1}$ uniformly in j . This bounds all q_j . Finally, the distance between a probability column and its starting basis vector is exactly twice the mass that left the diagonal. Taking the supremum over columns gives norm continuity at zero.

4 Partial theorem and proof

Theorem 1. *Let $(T_t)_{t \geq 0}$ be a triangular Markovian C_0 -semigroup on $\ell^1(\mathbb{N})$. Suppose that T_1 is bijective, and put*

$$M = \|T_1^{-1}\|.$$

Then, for every $t \geq 0$,

$$\|T_t - \text{id}\| \leq 2(1 - M^{-t}). \quad (2)$$

In particular the generator A of (T_t) is bounded and

$$\|A\| \leq 2 \log M. \quad (3)$$

Proof. Write $p_{ij}(t) = \langle T_t e_j, e_i^* \rangle$. Positivity and preservation of the ℓ^1 norm on the positive cone give

$$p_{ij}(t) \geq 0, \quad \sum_i p_{ij}(t) = 1.$$

Fix j . By the semigroup law,

$$p_{jj}(t+s) = \sum_{k \in \mathbb{N}} p_{jk}(t) p_{kj}(s).$$

Triangularity says that a nonzero factor $p_{kj}(s)$ requires $k \geq j$, whereas a nonzero factor $p_{jk}(t)$ requires $j \geq k$. Thus only $k = j$ contributes, and

$$p_{jj}(t+s) = p_{jj}(t) p_{jj}(s). \quad (4)$$

Strong continuity gives $p_{jj}(0) = 1$ and continuity in t . Consequently there is a number $q_j \geq 0$ such that

$$p_{jj}(t) = e^{-q_j t} \quad (t \geq 0). \quad (5)$$

We next use bijectivity at time one. For $j \in \mathbb{N}$, let

$$F_j = \text{span}\{e_1^*, \dots, e_j^*\} \subset \ell^\infty.$$

The subspace F_j is invariant under T_1^* . Indeed, if $f \in F_j$ and $k > j$, then

$$(T_1^* f)_k = \sum_i p_{ik}(1) f_i = 0,$$

because triangularity forces every potentially nonzero term to have $i \geq k > j$, while $f_i = 0$ there. In the displayed ordered basis, the finite-dimensional restriction $T_1^*|_{F_j}$ is triangular and has diagonal entries $p_{11}(1), \dots, p_{jj}(1)$. Therefore $p_{jj}(1)$ is an eigenvalue of T_1^* : choose a corresponding nonzero eigenvector $v \in F_j$.

Since T_1^* is invertible and $\|(T_1^*)^{-1}\| = \|T_1^{-1}\| = M$, the eigenvalue equation gives

$$\|v\| = \|(T_1^*)^{-1} p_{jj}(1) v\| \leq M p_{jj}(1) \|v\|.$$

Hence $p_{jj}(1) \geq M^{-1}$. By (5),

$$q_j \leq \log M \quad (j \in \mathbb{N}). \quad (6)$$

For each stochastic column, positivity and the column-sum identity yield

$$\begin{aligned} \|(T_t - \text{id})e_j\|_1 &= |p_{jj}(t) - 1| + \sum_{i \neq j} p_{ij}(t) \\ &= 2(1 - p_{jj}(t)). \end{aligned}$$

The norm of a matrix operator on ℓ^1 is the supremum of the ℓ^1 norms of its columns. Combining this fact with (5) and (6) gives

$$\|T_t - \text{id}\| = \sup_j 2(1 - e^{-q_j t}) \leq 2(1 - e^{-t \log M}) = 2(1 - M^{-t}),$$

which is (2).

In particular $\|T_t - \text{id}\| \rightarrow 0$ as $t \downarrow 0$. A C_0 -semigroup that is norm-continuous at zero has a bounded generator, and

$$A = \lim_{t \downarrow 0} \frac{T_t - \text{id}}{t}$$

in operator norm [2, Chapter I]. Dividing (2) by t and passing to the limit proves (3). $\square$

Corollary 2. *The Markov group conjecture is true for every countable-state process whose transition semigroup becomes triangular after one fixed relabeling of the states. If $\|T_1^{-1}\| = 1$, then $T_t = \text{id}$ for every $t \geq 0$.*

Proof. The first assertion is Theorem 1. For the second, (2) has right-hand side zero. $\square$

5 Verification notes

- No domain assumption on the basis vectors is made. The proof first obtains an operator-norm estimate and only then invokes the bounded-generator criterion.
- The eigenvalue argument occurs inside the genuinely invariant, finite-dimensional subspace F_j . It does not rely on a general claim that every diagonal entry of an infinite triangular operator lies in its spectrum.

- The triangular orientation is consistent throughout: $p_{ij} = 0$ for $i < j$ means transitions $j \rightarrow i$ only with $i \geq j$.
- The exact column identity uses both positivity and conservation of mass. It would fail for merely substochastic semigroups.
- No numerical or symbolic computation is used.

Verifier verdict: likely valid. The most useful human checks are the invariance of F_j under T_1^* and the equality $\|T_t - \text{id}\| = \sup_j 2(1 - p_{jj}(t))$.

6 Scope and novelty status

Cycles are the precise limitation: without triangularity, the (j, j) entry of T_{t+s} includes paths that leave j and later return, so (4) fails. The proof gives no control in that setting, and the full Markov group conjecture remains open here.

On 11 August 2026, the local lightweight indexes and parsed arXiv corpus were searched for the exact id, conjecture, and triangular, acyclic, and pure-birth variants. Bounded official-arXiv searches used the same terms and found no exact statement of this quantitative subcase. Novelty confidence is modest: the argument is elementary, and older Markov-chain literature outside arXiv may contain a comparable observation.

7 Human review recommendation

Verify the two points in the verdict. If accepted, retain this as a quantitative partial result for triangular countable-state processes. It must not be described as settling the full Markov group conjecture.

References

- [1] Jochen Glueck, *On the decoupled Markov group conjecture*, arXiv:2004.05995, 2020. <https://arxiv.org/abs/2004.05995>
- [2] Klaus-Jochen Engel and Rainer Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer, 2000.